

Smoothing Inequalities

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1 Warm up problems

Problem 1.1 Prove that for all non-negative real numbers a, b, c we have

$$a^3 + b^3 + c^3 - 3abc \geq 4(a-b)(b-c)(c-a).$$

Problem 1.2 (Jack Grafinkel, Crux) Prove that for all non-negative real numbers a, b, c the following inequality holds:

$$\frac{a}{\sqrt{a+b}} + \frac{b}{\sqrt{b+c}} + \frac{c}{\sqrt{c+a}} \leq \frac{5}{4}\sqrt{a+b+c}.$$

Problem 1.3 Let a, b, c be non-negative real numbers which are not all zero. Prove that

$$\frac{1}{2a^2 + bc} + \frac{1}{2b^2 + ca} + \frac{1}{2c^2 + ab} \geq \frac{8}{(a+b+c)^2}.$$

Problem 1.4 Let $a, b, c \geq 0$ such that $ab + bc + ca = 1$. Prove that

$$\frac{1}{(a+b)^2} + \frac{1}{(b+c)^2} + \frac{1}{(c+a)^2} \geq \frac{9}{4}.$$

2 Hard problems

CGMO 2011 Positive reals a, b, c, d satisfy $abcd = 1$. Prove that

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} + \frac{9}{a+b+c+d} \geq \frac{25}{4}.$$

ISL 2021 A5.

Let $n \geq 2$ be an integer, and let $a_1, a_2, \dots, a_n$ be positive real numbers such that

$$a_1 + a_2 + \dots + a_n = 1.$$

Prove that

$$\sum_{k=1}^n \frac{a_k}{1-a_k} (a_1 + a_2 + \dots + a_{k-1})^2 < \frac{1}{3}.$$

IMO 2021 P2 Show that for all real numbers $x_1, \dots, x_n$ the following inequality holds:

$$\sum_{i=1}^n \sum_{j=1}^n \sqrt{|x_i - x_j|} \leq \sum_{i=1}^n \sum_{j=1}^n \sqrt{|x_i + x_j|}.$$

IMO SL 2021 A7. Let $n \geq 1$ be an integer, and let $x_0, x_1, \dots, x_{n+1}$ be $n+2$ non-negative real numbers that satisfy $x_i x_{i+1} - x_{i-1}^2 \geq 1$ for all $i = 1, 2, \dots, n$. Show that

$$x_0 + x_1 + \dots + x_n + x_{n+1} > \left(\frac{2n}{3}\right)^{3/2}.$$

ISL 2020 A7. Let n and k be positive integers. Prove that for $a_1, \dots, a_n \in [1, 2^k]$ one has

$$\sum_{i=1}^n \frac{a_i}{\sqrt{a_1^2 + \dots + a_i^2}} \leq 4\sqrt{kn}.$$